HANGOO KANG

AI Researcher — LLM Post-Training, Agentic AI, Distributed Training
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EDUCATION

Stanford University *Aug 2025 — May 2027*
M.S. in Computer Science (AI track) Stanford, CA
Advised by Profs. Azalia Mirhoseini and Yejin Choi.

University of Illinois Urbana–Champaign *Aug 2021 — May 2025*
B.S. in Computer Science | GPA: 3.96 / 4.00 Urbana, IL
Dean's List: Spring 2022, Spring 2023, Fall 2023, Spring 2024.

SELECTED PUBLICATIONS ** equal contribution*

[1] TRACE: Capability-Targeted Agentic Training.
Hangoo Kang*, Tarun Suresh*, Jon Saad-Falcon, Azalia Mirhoseini.
AIWILD @ ICML 2026, under review for full conference. paper · code

[2] Forty Shades of Blue: Quality-Diversity Alignment via Mode-Conditioned Reinforcement Learning.
Hangoo Kang*, Jiayi Yuan*, James Jihao Liu*, Yejin Choi, Vikram Iyer, Liwei Jiang, Natasha Jaques.
GenAICreativity @ ICML 2026, under review for full conference.

[3] TRAP: Targeted Redirecting of Agentic Preferences.
Hangoo Kang*, Jehyeok Yeon*, Gagandeep Singh.
NeurIPS 2025 paper · code

[4] Learning a Pessimistic Reward in RLHF: KL Regularization is Not Necessary.
Hangoo Kang*, Yinglun Xu*, Tarun Suresh, Yuxuan Wan, Gagandeep Singh.
ARLET @ NeurIPS 2025 paper

[5] Stochastic Monkeys at Play: Random Augmentations Cheaply Break LLM Safety Alignment.
Jason Vega, **Hangoo Kang***, Junsheng Huang*, Gaokai Zhang*, Minjia Zhang, Gagandeep Singh.
Under review. paper · code

[6] SynCode: LLM Generation with Grammar Augmentation.
Shubham Ugare, Tarun Suresh, **Hangoo Kang**, Sasa Misailovic, Gagandeep Singh.
TMLR 2025 paper · code

RESEARCH EXPERIENCE

Stanford University — Scaling Intelligence Lab *Aug 2025 — Present*
Graduate Researcher · Advised by Profs. Azalia Mirhoseini Stanford, CA

- Lead author of *TRACE* (AIWILD @ ICML 2026): end-to-end system for capability-targeted agent self-improvement that contrasts successful and failed trajectories, synthesizes per-capability training environments, and trains LoRA adapters via GRPO, achieving +14.1 pts on τ^2-Bench and +9.2 pts over GRPO at matched rollout budget.
- Co-author on PyTorch / Meta Superintelligence Lab production blog releases on *torchforge* (PyTorch-native scalable RL post-training): Introducing TorchForge and Scalable RL with TorchForge and Weaver; contributor to the open-source meta-pytorch/torchforge repository.
- Drove the Forge × Weaver verifier integration validated on a 512-GPU CoreWeave cluster, reaching 4× faster RL step times, >90% job completion, and matching o3-mini on MATH500 / GPQA / MMLU-Pro without human annotations.
- Co-leading the universal verifier project for agentic tasks to avoid domain specific post training with low compute requirement.

Stanford University — xLab *Aug 2025 — Present*
Graduate Researcher · Advised by Prof. Yejin Choi (Stanford) & Prof. Natasha Jaques (UW) Stanford, CA

- Co-first author on *MODA*, investigating model collapse and inter-model homogeneity (the "Artificial Hivemind" effect) in frontier LMs (LLaMA-3, Gemini-2.5, GPT-4o), and designing multi-agent RL training to recover diverse, human-like generation distributions.
- Building an automated synthetic data generation framework for LLM alignment in out-of-distribution and underspecified domains.

UIUC — FOCAL Lab & ARC Group *Spring 2023 — Spring 2025*
Undergraduate Researcher · Advised by Profs. Gagandeep Singh, Sasa Misailovic, Minjia Zhang Urbana, IL

- First-author of *TRAP* (NeurIPS 2025): designed an optimization-driven adversarial attack that embeds imperceptible prompts in images to reliably steer GUI-based AI agents toward attacker-chosen targets.
- Led the *Pessimistic Reward Trainer* project (ARLET @ NeurIPS 2025), addressing reward hacking in RLHF without KL regularization.
- Co-developed *SynCode* (TMLR 2025), a grammar-guided decoding framework that enforces syntactic correctness in LLM generation.
- Studied random character-based prompt augmentation attacks on LLM safety alignment (paper under review).

UIUC — Crop Science Research Lab *Spring 2024 — Spring 2025*

Undergraduate Researcher · Advised by Dr. Jinwook Kim Urbana, IL

- Developed predictive models for final grain biomass as a function of nutrient supply, deployed on a cleaned real-world agricultural dataset.

INDUSTRY COLLABORATIONS & OPEN SOURCE

Meta Superintelligence Lab (PyTorch team) *Aug 2025 — Feb 2026*

Collaborated on *torchforge*, a PyTorch-native scalable RL library; co-author on two official PyTorch blog releases and contributor to the open-sourced meta-pytorch/torchforge repo. Helped scale RLVR-style training to a 512-GPU CoreWeave cluster.

CoreWeave *Aug 2025 — Feb 2026*

Joint validation of distributed RL workloads on Slurm-on-Kubernetes (SUNK) with NVIDIA Quantum InfiniBand.

SELECTED PROJECTS

Heart Disease Prediction, Hybrid CNN + BERT *Aug 2023 — Dec 2023*

Led 6 engineers; fused image (CNN) and tabular/text (BERT) signals to improve classification accuracy by +32% over the numeric-only baseline.

AutoTA, Video-to-Notes Chrome Extension *Aug 2023 — Dec 2023*

Led 5 engineers to build a lecture-video summarization pipeline (word2vec + PEGASUS) shipped as a Chrome extension.

Build My Portfolio, RL for Asset Allocation *Jan 2023 — Jul 2023*

Built a stock-trading RL environment with self-improving agents; Transformer/LSTM ticker clustering inspired by Markowitz portfolio theory.

MindMe, Personality-Classifying Conversational Agent *Aug 2021 — Jan 2023*

Built a TensorFlow chatbot that infers user personality from open-ended dialogue by combining a BERT decoding head, a sentiment-analysis layer, and a CNN text-classification pipeline into a single inference stack.

ACADEMIC SERVICE

- **Conference Reviewer:** ICLR 2026, ICML 2026, COLM 2026, NeurIPS 2025.

TEACHING & LEADERSHIP

President, Codable @ UIUC *Spring 2024 — Spring 2025*

Led 50+ members per semester; built the org's NodeJS website used by 150+ students; ran algorithm and tech-interview workshops in partnership with CodeTree.

Course Assistant, CS 124 (Intro to CS), UIUC *Fall 2022*

Weekly code reviews, semester-long project mentorship, and a four-week supplementary lecture series.

Peer Tutor, MATH 241 (Calculus III), UIUC *Fall 2022*

Raised group average from 78.5% to 94.0% (+15.5 pts) across vector-calculus topics.

SELECTED AWARDS

UIUC Datathon, Honorable Mention (top 10 of 200+ teams) for fraud / anomaly / credit-scoring pipeline (autoencoder + BERT, +12% anomaly-detection lift). *2025*

Illinois Engineering Outstanding Scholarship ($8000 per year) *2023, 2024*

Illinois Engineering Achievement Scholarship ($8000 per year) *2023, 2024*

HackIllinois, 2nd Place, Beginner Division. *2022*

TECHNICAL SKILLS

Research areas	LLM post-training (RLHF, RLVR), agentic RL, adversarial robustness, verifier design, MLSys
ML / RL stack	PyTorch, torchforge, TorchTitan, vLLM, Monarch, TRL, VeRL, DeepSpeed, TensorFlow, JAX
Infra & tooling	CUDA, distributed training (FSDP, tensor/pipeline parallelism), Slurm, Kubernetes, AWS, Docker, Git.